

1.2 Algebra & Functions

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

1. Indices and Surds

The rules of indices (exponents) underpin nearly every algebraic manipulation in the A-Level course. They hold for any real base $a > 0$ and any real exponents m, n ; for integer exponents they hold for any non-zero base.

Laws of Indices

Law	Statement	Example
Product	$a^m \cdot a^n = a^{m+n}$	$x^3 \cdot x^5 = x^8$
Quotient	$a^m / a^n = a^{m-n}$	$x^7 / x^2 = x^5$
Power of a power	$(a^m)^n = a^{mn}$	$(x^4)^3 = x^{12}$
Power of a product	$(ab)^n = a^n b^n$	$(2x)^3 = 8x^3$
Power of a quotient	$(a/b)^n = a^n / b^n$	$(x/3)^2 = x^2 / 9$
Zero index	$a^0 = 1$ (for $a \neq 0$)	$7^0 = 1$
Negative index	$a^{-n} = 1/a^n$	$x^{-3} = 1/x^3$
Fractional index	$a^{1/n} = \sqrt[n]{a}, a^{m/n} = \sqrt[n]{a^m}$	$8^{2/3} = (\sqrt[3]{8})^2 = 4$

Surd. An irrational number expressed in the form $\sqrt[n]{a}$ where a is a positive integer and the n -th root is not itself a rational number. Common surds at A-level are $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{10}$.

Simplifying Surds

Key rules: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$, valid for $a, b \geq 0$. To simplify $\sqrt{N}$, factor out the largest perfect-square divisor of N .

Worked Example 1 — Simplifying surds

Simplify (a) $\sqrt{50}$, (b) $\sqrt{72} - \sqrt{18}$, (c) $\sqrt{12} \cdot \sqrt{27}$.

// (a) factor out the largest square

$$1. \quad \sqrt{50} = \sqrt{(25 \cdot 2)} = \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2}.$$

// (b) simplify each surd then combine

$$2. \quad \sqrt{72} = \sqrt{(36 \cdot 2)} = 6\sqrt{2}$$

$$3. \quad \sqrt{18} = \sqrt{(9 \cdot 2)} = 3\sqrt{2}$$

$$4. \quad \sqrt{72} - \sqrt{18} = 6\sqrt{2} - 3\sqrt{2} = 3\sqrt{2}.$$

// (c) combine under one radical

$$5. \quad \sqrt{12} \cdot \sqrt{27} = \sqrt{(12 \cdot 27)} = \sqrt{324} = 18.$$

Rationalising the Denominator

A denominator containing a surd is "rationalised" by multiplying numerator and denominator by a convenient surd expression to remove the irrational part.

- $\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$ — multiply top and bottom by $\sqrt{a}$.
- $\frac{1}{a + \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$ — multiply by the *conjugate* $a - \sqrt{b}$.
- $\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{\sqrt{a} - \sqrt{b}}{a - b}$.

Worked Example 2 — Rationalising

Express $\frac{5}{3 - \sqrt{2}}$ in the form $p + q\sqrt{2}$.

1. Multiply by conjugate $(3 + \sqrt{2})/(3 + \sqrt{2})$:
2. $5/(3 - \sqrt{2}) \cdot (3 + \sqrt{2})/(3 + \sqrt{2})$
3. $= 5(3 + \sqrt{2}) / (3^2 - (\sqrt{2})^2)$
4. $= (15 + 5\sqrt{2}) / (9 - 2)$
5. $= (15 + 5\sqrt{2}) / 7$
6. $= 15/7 + (5/7)\sqrt{2}$. // so $p = 15/7$, $q = 5/7$

Exam tip. OCR examiners routinely deduct marks when students leave a surd in the denominator. Always rationalise as the final step unless the question explicitly leaves an alternative form. If the denominator is $a \pm \sqrt{b}$, multiply by $a \mp \sqrt{b}$ — the difference of two squares does the work for you.

2. Quadratics

A quadratic has the form $ax^2 + bx + c$ with $a \neq 0$. Three standard techniques solve $ax^2 + bx + c = 0$: factorising, completing the square, and the quadratic formula.

Quadratic formula. The solutions of $ax^2 + bx + c = 0$ (with $a \neq 0$) are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Discriminant. The quantity $\Delta = b^2 - 4ac$. Its sign determines the number and type of real roots and the way the parabola meets the x -axis.

Discriminant Cases

Sign of Δ	Roots	Graph behaviour
$\Delta > 0$	Two distinct real roots	Parabola crosses x -axis at two points
$\Delta = 0$	One repeated real root	Parabola touches x -axis (tangent) at vertex

$\Delta < 0$	No real roots (complex conjugate pair)	Parabola does not meet x -axis
Δ is a perfect square	Rational roots — factorises over $\mathbb{Q}$	Easiest case to factor by inspection

Completing the Square

Any quadratic $ax^2 + bx + c$ can be written in *vertex form*: $a(x - h)^2 + k$, where $h = -\frac{b}{2a}$ and $k = c - \frac{b^2}{4a}$. The vertex of $y = ax^2 + bx + c$ sits at (h, k) ; the minimum (if $a > 0$) or maximum (if $a < 0$) value is k .

Worked Example 3 — Solve $2x^2 - 5x + 3 = 0$ three ways

// Method A — factorising

- $2x^2 - 5x + 3 = 0$
- Find two numbers that multiply to $2 \cdot 3 = 6$ and sum to -5 : -2 and -3 .
- Split middle term: $2x^2 - 2x - 3x + 3 = 0$
- $2x(x - 1) - 3(x - 1) = 0$
- $(2x - 3)(x - 1) = 0$
- $\Rightarrow x = 3/2$ or $x = 1$.

// Method B — quadratic formula

- $a = 2$, $b = -5$, $c = 3$, $\Delta = 25 - 24 = 1$.
- $x = (5 \pm \sqrt{1}) / 4 = (5 \pm 1)/4$.
- $\Rightarrow x = 6/4 = 3/2$, or $x = 4/4 = 1$.

// Method C — completing the square

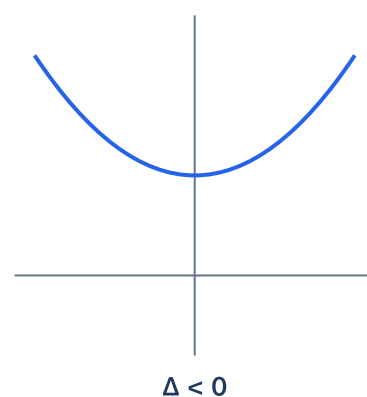
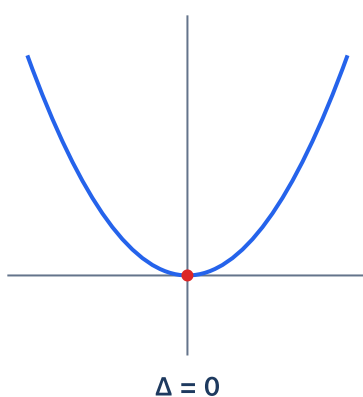
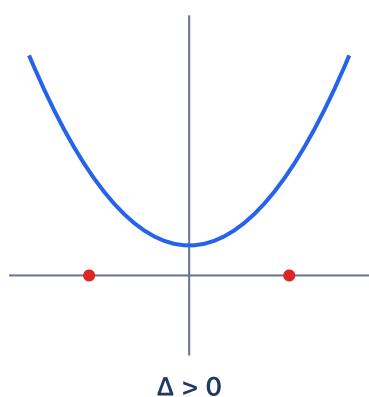
- $2x^2 - 5x + 3 = 2(x^2 - 5x/2) + 3$
- $= 2[(x - 5/4)^2 - 25/16] + 3$
- $= 2(x - 5/4)^2 - 25/8 + 3$
- $= 2(x - 5/4)^2 - 1/8$.
- Set to 0: $2(x - 5/4)^2 = 1/8$
- $(x - 5/4)^2 = 1/16$
- $x - 5/4 = \pm 1/4$
- $\Rightarrow x = 5/4 + 1/4 = 3/2$, or $x = 5/4 - 1/4 = 1$.

The vertex form also tells us the minimum value of the quadratic is $-1/8$, attained at $x = 5/4$.

Exam tip — when to use which method. Try factorising first: it is fastest when Δ is a small perfect square. If factorising is not obvious in ~15 seconds, switch to the formula. Use completing the square when the question asks for the vertex, the minimum/maximum value, the range, or to prove that the quadratic is always positive/negative.

Sketching Quadratics

To sketch $y = ax^2 + bx + c$: (i) direction of opening — up if $a > 0$, down if $a < 0$; (ii) y -intercept $(0, c)$; (iii) x -intercepts from the roots (if real); (iv) vertex from completing the square.



Worked Example 4 — Quadratic word problem

A rectangular field is 5 m longer than it is wide and has area 84 m². Find its dimensions.

Let width = x . Then length = $x + 5$ and $x(x + 5) = 84$, i.e. $x^2 + 5x - 84 = 0$.

Discriminant = $25 + 336 = 361 = 19^2$. Roots: $x = (-5 \pm 19)/2$, giving $x = 7$ or $x = -12$.

Width must be positive, so $x = 7$ m, length = 12 m.

3. Simultaneous Equations

Simultaneous equations have a solution set consisting of every pair (or n -tuple) that satisfies both (all) equations. Geometrically each equation is a curve; solutions are intersection points.

Linear / Linear Systems

Two methods: **substitution** (solve one equation for a variable, substitute into the other) and **elimination** (scale the equations and add/subtract to cancel a variable). Exactly one solution iff the two lines are not parallel.

Worked Example 5 — Linear/linear

Solve $3x + 2y = 16$ and $x - y = 2$.

1. From (2): $x = y + 2$.
2. Substitute: $3(y + 2) + 2y = 16$
3. $3y + 6 + 2y = 16$
4. $5y = 10$
5. $y = 2, \quad x = 4$.
6. Check: $3(4) + 2(2) = 16 \checkmark$ and $4 - 2 = 2 \checkmark$.

Geometrically, the lines $3x + 2y = 16$ and $x - y = 2$ intersect at $(4, 2)$.

Linear / Quadratic Systems

Substitute the linear equation into the quadratic to reduce to a single quadratic in one variable. Possibilities: two solutions (line is a secant — cuts the curve), one solution (line is tangent), no solutions (line misses the curve). The discriminant of the resulting quadratic tells you which case you are in.

Worked Example 6 — Linear/quadratic

Solve $y = x^2 - 2x - 3$ and $y = x - 3$. Interpret the result geometrically.

1. $x^2 - 2x - 3 = x - 3$
2. $x^2 - 3x = 0$
3. $x(x - 3) = 0$
4. $x = 0$ or $x = 3$.

5. Corresponding y : $y = 0 - 3 = -3$; $y = 3 - 3 = 0$.
6. $\therefore$ intersection points: $(0, -3)$ and $(3, 0)$.

The line $y = x - 3$ crosses the parabola $y = x^2 - 2x - 3$ at exactly these two points (two real solutions $\Rightarrow$ line is a secant).

4. Inequalities

When manipulating inequalities, remember: multiplying or dividing both sides by a *negative* number **reverses** the inequality.

Linear Inequalities

Treat like a linear equation, but reverse the sign when multiplying/dividing by negatives.

Example: $3 - 2x > 7 \Rightarrow -2x > 4 \Rightarrow x < -2$.

Quadratic Inequalities

Solve $ax^2 + bx + c > 0$ (or $<$, $\leq$, $\geq$) in three steps:

1. Find the roots α, β of $ax^2 + bx + c = 0$.
2. Sketch the parabola (or draw a sign diagram) noting whether it opens up ($a > 0$) or down ($a < 0$).
3. Read off the x -values where the sketch lies above/below the x -axis as required.

Worked Example 7 — Quadratic inequality

Solve $x^2 - x - 6 \leq 0$.

1. Roots of $x^2 - x - 6 = 0$: $(x - 3)(x + 2) = 0 \Rightarrow x = -2, x = 3$.
2. Parabola opens upwards (coefficient of x^2 is $+1$).
3. $y \leq 0$ between the roots.
4. $\therefore$ solution: $-2 \leq x \leq 3$.

Exam tip — sign diagram for a quadratic inequality. Always sketch the parabola, even a rough one. A common error is to write $x < -2$ or $x > 3$ for $x^2 - x - 6 \leq 0$ — that is the solution to the > 0 version. The sketch prevents this inversion of the answer. Remember: *between the roots* for ≤ 0 (upward parabola), *outside the roots* for ≥ 0 .

Modulus Inequalities

The modulus $|x|$ means the non-negative value of x . Useful equivalences:

- $|x| = a$ ($a \geq 0$) iff $x = a$ or $x = -a$.
- $|x| \leq a$ iff $-a \leq x \leq a$.
- $|x| \geq a$ iff $x \leq -a$ or $x \geq a$.

Example: $|2x - 1| < 5 \Rightarrow -5 < 2x - 1 < 5 \Rightarrow -4 < 2x < 6 \Rightarrow -2 < x < 3$.

5. Polynomials

Polynomial. An expression of the form $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_n$ are real constants with $a_n \neq 0$. The *degree* is n and the *leading coefficient* is a_n .

Polynomial Division

To divide polynomial $f(x)$ by $g(x)$ ($\deg g \leq \deg f$), use long division:

- Divide leading term of f by leading term of g — this is the next term of the quotient.
- Multiply $g(x)$ by that term, subtract from $f(x)$.
- Repeat on the remainder until the remainder has degree $< \deg g$.

Worked Example 8 — Polynomial long division

Divide $2x^3 - 3x^2 - 8x + 12$ by $x - 2$.

$$2x^2 + x - 6 \quad // \text{ quotient}$$

$$\begin{array}{r}
 x - 2 \) \ 2x^3 - 3x^2 - 8x + 12 \\
 \underline{2x^3 - 4x^2} \\
 x^2 - 8x \\
 \underline{x^2 - 2x} \\
 -6x + 12 \\
 \underline{-6x + 12} \\
 0
 \end{array}$$

// subtract: $2x^2 \cdot (x-2)$

// subtract: $x \cdot (x-2)$

// subtract: $-6 \cdot (x-2)$

// remainder

So $2x^3 - 3x^2 - 8x + 12 = (x - 2)(2x^2 + x - 6)$, with remainder 0.

Factor and Remainder Theorems

Factor theorem. For a polynomial $f(x)$ and a constant a : $(x - a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.

Remainder theorem. When $f(x)$ is divided by $(x - a)$, the remainder equals $f(a)$. (The factor theorem is the special case where this remainder is 0.)

Worked Example 9 — Factor theorem applied

Factorise $x^3 + 2x^2 - 5x - 6$ completely.

1. Try small integers dividing the constant term -6 : $\pm 1, \pm 2, \pm 3, \pm 6$.
2. $f(-1) = -1 + 2 + 5 - 6 = 0$. // so $(x + 1)$ is a factor
3. Divide $x^3 + 2x^2 - 5x - 6$ by $(x + 1)$:
quotient = $x^2 + x - 6$.
4. Factor the quadratic: $x^2 + x - 6 = (x + 3)(x - 2)$.
5. $\therefore x^3 + 2x^2 - 5x - 6 = (x + 1)(x + 3)(x - 2)$.
6. Roots: $x = -3, -1, 2$.

Exam tip — searching for roots. For a polynomial with integer coefficients and leading coefficient 1, any rational root must be an integer dividing the constant term. For leading coefficient a_n , any rational root p/q satisfies $p \mid a_0$ and $q \mid a_n$. Start with ± 1 , then ± 2 , before trying larger divisors.

Sketching Cubics and Quartics

Given a cubic in factorised form $y = a(x - r_1)(x - r_2)(x - r_3)$:

- Three distinct real roots — the curve crosses the x -axis three times.
- A repeated root $(x - r)^2$ — the curve *touches* (is tangent to) the x -axis at $x = r$.
- A triple root $(x - r)^3$ — the curve has a point of inflection on the x -axis at $x = r$.
- As $x \rightarrow +\infty$: $y \rightarrow +\infty$ if $a > 0$; $y \rightarrow -\infty$ if $a < 0$. Reverse for $x \rightarrow -\infty$.

Analogous rules apply to quartics: leading coefficient positive gives "up then up" end behaviour; leading coefficient negative gives "down then down".

6. Graphs of Functions — Transformations

Starting from the graph of $y = f(x)$, a sequence of standard transformations produces related graphs. The four families are *translation*, *stretch*, *reflection*, and combinations thereof.

Transformation Summary

New function	Effect on graph of $y = f(x)$	Worded description
$f(x) + a$	Translate vertically	Shift up by a (down if $a < 0$)
$f(x - a)$	Translate horizontally	Shift right by a (left if $a < 0$)
$a \cdot f(x)$	Vertical stretch	Scale y -coordinates by factor a
$f(ax)$	Horizontal stretch	Scale x -coordinates by factor $1/a$
$-f(x)$	Reflect in x -axis	Flip vertically
$f(-x)$	Reflect in y -axis	Flip horizontally

$ f(x) $	Reflect negative parts upwards	Any part below the x -axis is flipped above
$f(x)$	Right half mirrored to left	Keep $x \geq 0$ part; reflect it in the y -axis

Exam tip — direction of horizontal shifts. Horizontal transformations behave *opposite* to what the sign suggests. $f(x - 3)$ shifts the graph **right** by 3 (because the x -value that used to give $f(0)$ now needs $x = 3$). Similarly $f(2x)$ **compresses** horizontally by factor 2; $f(x/2)$ stretches by factor 2. Many students get the sign wrong under pressure — practice this until it's automatic.

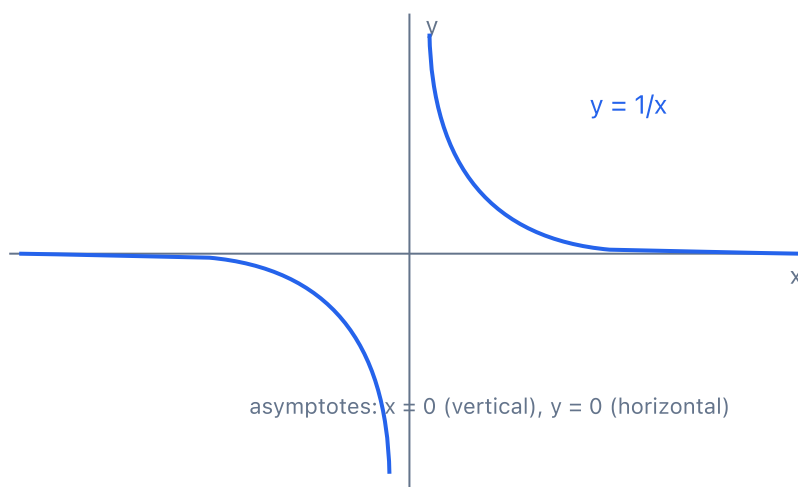
Combining Transformations

Order matters. For $y = af(b(x - c)) + d$: the inside ($b(x - c)$) acts first on x — horizontal stretch by $1/b$, then translation right by c ; the outside ($a \cdot +d$) acts next on y — vertical stretch by a , then translation up by d .

7. Rational Functions and Asymptotes

A rational function is a quotient of two polynomials, $y = P(x)/Q(x)$. The standard A-level example is $y = 1/x$.

- **Vertical asymptote:** $x = a$ where $Q(a) = 0$ but $P(a) \neq 0$. The function tends to $\pm\infty$ as $x \rightarrow a$.
- **Horizontal asymptote:** behaviour as $x \rightarrow \pm\infty$. If $\deg P < \deg Q$, the horizontal asymptote is $y = 0$; if $\deg P = \deg Q$, it is $y = \text{ratio of leading coefficients}$.



Translations apply as usual: $y = 1/(x - 1)$ shifts the graph of $y = 1/x$ right by 1, giving a vertical asymptote at $x = 1$. $y = 1/x + 2$ shifts it up by 2, giving a horizontal asymptote at $y = 2$.

8. Partial Fractions

Partial fractions. A technique that decomposes a proper rational expression $P(x)/Q(x)$ (with $\deg P < \deg Q$) into a sum of simpler fractions whose denominators are the factors of $Q(x)$.

If $\deg P \geq \deg Q$, first perform polynomial division to obtain a polynomial plus a proper rational remainder, then decompose the remainder.

Setup by Denominator Type

Factor in $Q(x)$	Corresponding partial fraction(s)
Distinct linear $(x - a)$	$\frac{A}{x - a}$
Repeated linear $(x - a)^2$	$\frac{A}{x - a} + \frac{B}{(x - a)^2}$
Repeated linear $(x - a)^3$	$\frac{A}{x - a} + \frac{B}{(x - a)^2} + \frac{C}{(x - a)^3}$
Irreducible quadratic $(x^2 + bx + c), \Delta < 0$	$\frac{Ax + B}{x^2 + bx + c}$

Worked Example 10 — Distinct linear factors

Express $\frac{3x}{(x - 1)(x + 2)}$ in partial fractions.

- Set up: $3x / [(x - 1)(x + 2)] = A/(x - 1) + B/(x + 2)$.
- Multiply through by $(x - 1)(x + 2)$:
- $3x = A(x + 2) + B(x - 1)$.
- Substitute $x = 1$: $3(1) = A(3) + B(0) \Rightarrow A = 1$.
- Substitute $x = -2$: $3(-2) = A(0) + B(-3) \Rightarrow B = 2$.

$$6. \quad \therefore 3x / [(x-1)(x+2)] = 1/(x-1) + 2/(x+2).$$

$$7. \quad \text{Check: } 1/(x-1) + 2/(x+2) = [(x+2) + 2(x-1)] / [(x-1)(x+2)] = 3x / [(x-1)(x+2)]$$

Worked Example 11 — Repeated linear factor

Express $\frac{2x+1}{(x-1)(x+2)^2}$ in partial fractions.

$$1. \quad \text{Set up: } (2x+1) / [(x-1)(x+2)^2]$$

$$= A/(x-1) + B/(x+2) + C/(x+2)^2.$$

$$2. \quad \text{Multiply up: } 2x+1 = A(x+2)^2 + B(x-1)(x+2) + C(x-1).$$

$$3. \quad x=1: \quad 3 = A(9) + 0 + 0 \quad \Rightarrow \quad A = 1/3.$$

$$4. \quad x=-2: \quad -3 = 0 + 0 + C(-3) \quad \Rightarrow \quad C = 1.$$

$$5. \quad \text{Compare } x^2 \text{ coefficients: } 0 = A + B \quad \Rightarrow \quad B = -1/3.$$

$$6. \quad \therefore (2x+1)/[(x-1)(x+2)^2]$$

$$= 1/[3(x-1)] - 1/[3(x+2)] + 1/(x+2)^2.$$

Exam tip — setting up partial fractions. The most common error is forgetting the repeated-factor rule. $(x-a)^2$ contributes *two* terms: $A/(x-a)$ and $B/(x-a)^2$. Missing the first of these produces a system with no solution — if you find your simultaneous equations are inconsistent, check the setup before hunting for arithmetic slips.

9. Functions

Function. A rule $f : A \rightarrow B$ that assigns to each element x of a set A (the *domain*) exactly one element $f(x)$ of B . The set of actual output values $\{f(x) : x \in A\}$ is the *range* (a subset of B , the codomain).

Key Terminology

- **One-to-one (injective).** Different inputs give different outputs: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Graph passes the *horizontal line test* — every horizontal line meets the graph in at most one point.

- **Many-to-one.** Two or more inputs map to the same output. E.g. $f(x) = x^2$ on $\mathbb{R}$ is many-to-one since $f(2) = f(-2) = 4$.
- **Onto (surjective).** Every element of the codomain is hit: range = codomain.
- **Bijective.** Both injective and surjective — there is a perfect pairing between domain and codomain. A function is invertible if and only if it is bijective (or, by restricting the codomain to the range, merely injective).

Inverse function. If $f : A \rightarrow B$ is one-to-one, its inverse $f^{-1} : \text{Range}(f) \rightarrow A$ is defined by $f^{-1}(y) = x$ iff $f(x) = y$. The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$. Domain of $f^{-1} = \text{Range of } f$; Range of $f^{-1} = \text{Domain of } f$.

Finding an Inverse Algebraically

1. Write $y = f(x)$.
2. Swap x and y (or rearrange to make x the subject, then relabel).
3. Solve for the new y ; this gives $f^{-1}(x)$.
4. State the domain of f^{-1} — it is the range of f .

Worked Example 12 — Inverse of a rational function

Let $f(x) = \frac{2x+3}{x-1}$, defined for $x \in \mathbb{R}$ with $x \neq 1$. Find $f^{-1}(x)$ and state its domain.

1. Let $y = (2x + 3)/(x - 1)$.
2. Multiply up: $y(x - 1) = 2x + 3$
3. $yx - y = 2x + 3$
4. $yx - 2x = y + 3$
5. $x(y - 2) = y + 3$
6. $x = (y + 3)/(y - 2)$.
7. Swap variables: $f^{-1}(x) = (x + 3)/(x - 2)$.
8. Domain of $f^{-1} = \text{Range of } f$.
9. Since $f(x) = 2 + 5/(x - 1)$, f takes every real value except 2.
10. $\therefore$ Domain of f^{-1} is $\{x \in \mathbb{R} : x \neq 2\}$.

Note the domain of f^{-1} is *not* simply "all reals" — the value $x = 2$ makes the denominator zero. Stating the domain is expected on every inverse-function question.

Exam tip — always state the domain. OCR mark schemes routinely allocate a dedicated mark for the domain of f^{-1} . The domain of f^{-1} is the range of f ; find it by (i) sketching $y = f(x)$, (ii) completing the square, or (iii) finding values f cannot attain. Forgetting this costs a mark almost every time.

Composition of Functions

The composition $(f \circ g)(x) = f(g(x))$ means: apply g first, then apply f to the result. In general $f \circ g \neq g \circ f$. The domain of $f \circ g$ is $\{x \in \text{Domain}(g) : g(x) \in \text{Domain}(f)\}$.

Example: if $f(x) = x + 2$ and $g(x) = x^2$, then $(f \circ g)(x) = x^2 + 2$ whereas $(g \circ f)(x) = (x + 2)^2 = x^2 + 4x + 4$.

Self-inverse property: $f \circ f^{-1}(x) = x$ and $f^{-1} \circ f(x) = x$ on the appropriate domains. This is often used as a verification step.

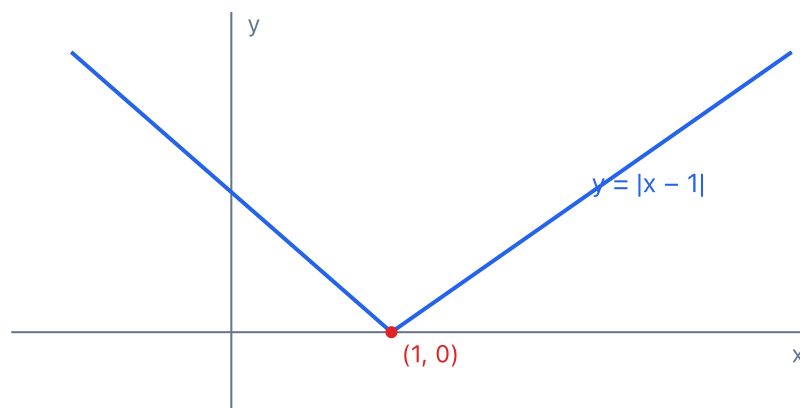
10. The Modulus Function

Modulus (absolute value). $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$. Equivalently $|x| = \sqrt{x^2}$.

Geometrically, $|x|$ is the distance from x to 0 on the number line; $|x - a|$ is the distance from x to a .

Graph of $y = |f(x)|$

Sketch $y = f(x)$, then reflect any part below the x -axis in the x -axis.



Solving Modulus Equations

Two standard approaches to $|f(x)| = g(x)$:

- **Case method.** $|f(x)| = g(x) \Leftrightarrow f(x) = g(x)$ or $f(x) = -g(x)$, with a final check that $g(x) \geq 0$ at each candidate.
- **Graphical method.** Sketch both curves; intersection points are the solutions.

Worked Example 13 — Modulus equation

Solve $|2x - 3| = x + 4$.

// Case 1: $2x - 3 \geq 0$, i.e. $x \geq 3/2$

1. $2x - 3 = x + 4 \Rightarrow x = 7$. Check $x \geq 3/2$ ✓ and RHS = $11 > 0$ ✓.

// Case 2: $2x - 3 < 0$, i.e. $x < 3/2$

2. $-(2x - 3) = x + 4 \Rightarrow -2x + 3 = x + 4 \Rightarrow -3x = 1 \Rightarrow x = -1/3$.

3. Check: $x < 3/2$ ✓, and RHS = $-1/3 + 4 = 11/3 > 0$ ✓.

4. $\therefore$ solutions: $x = -1/3$ or $x = 7$.

For inequalities $|f(x)| \leq g(x)$ equivalent to $-g(x) \leq f(x) \leq g(x)$ (when $g(x) \geq 0$). For $|f(x)| \geq g(x)$, the solution is $f(x) \leq -g(x)$ or $f(x) \geq g(x)$.

11. Curve Sketching — Putting It All Together

A complete sketch of $y = f(x)$ should show:

1. **Intercepts** — values where $x = 0$ (the y -intercept) and $f(x) = 0$ (the x -intercepts).
2. **Stationary points** — solve $f'(x) = 0$; classify as max/min/inflection.
3. **Asymptotes** — vertical where the function is undefined, horizontal from end behaviour.
4. **End behaviour** — as $x \rightarrow \pm\infty$, what does $f(x)$ do?
5. **Symmetries** — is f even ($f(-x) = f(x)$) or odd ($f(-x) = -f(x)$)?

Worked Example 14 — Sketch $y = x^3 - 3x$

1. Intercepts: $y(0) = 0$. Roots of $x^3 - 3x = x(x^2 - 3) = 0 \Rightarrow x = 0, \pm\sqrt{3}$.
2. Stationary points: $y' = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$.
3. At $x = 1$: $y = 1 - 3 = -2$ (local min, $y'' = 6x = 6 > 0$).
4. At $x = -1$: $y = -1 + 3 = 2$ (local max, $y'' = -6 < 0$).
5. End behaviour: $x \rightarrow +\infty \Rightarrow y \rightarrow +\infty$; $x \rightarrow -\infty \Rightarrow y \rightarrow -\infty$.
6. Symmetry: $f(-x) = -x^3 + 3x = -f(x)$ — odd, rotationally symmetric about origin.

12. Binomial Expansion — Positive Integer Index

Binomial coefficient. For non-negative integers $n \geq r$, $\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$. It counts the number of ways of choosing r objects from n , and equals the coefficient of x^r in $(1+x)^n$.

The **Binomial Theorem** (for $n \in \mathbb{Z}^+$):

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + b^n.$$

Pascal's Triangle

Row n of Pascal's triangle gives $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$. Each entry is the sum of the two entries above it.

$$\begin{array}{lcl} n = 0: & & 1 \\ n = 1: & 1 & 1 \end{array}$$

$n = 2:$		1	2	1			
$n = 3:$		1	3	3	1		
$n = 4:$		1	4	6	4	1	
$n = 5:$		1	5	10	10	5	1
$n = 6:$	1	6	15	20	15	6	1

Worked Example 15 — Expand $(1 + 2x)^5$

1. Use $(a + b)^5$ with $a = 1$, $b = 2x$. Coefficients from row 5: 1, 5, 10, 10, 5,
2. $(1 + 2x)^5$
 $= 1 + 5(2x) + 10(2x)^2 + 10(2x)^3 + 5(2x)^4 + (2x)^5$
 $= 1 + 10x + 40x^2 + 80x^3 + 80x^4 + 32x^5.$
3. Check constant term ($x=0$): 1 ✓. Check sum of coefficients ($x=1$): $3^5 = 243$

Use in approximation. If x is small, $(1 + 2x)^5 \approx 1 + 10x + 40x^2$, because higher powers of x are negligible. This lets you approximate numerical values such as 1.02^5 by setting $x = 0.01$: $1 + 10(0.01) + 40(0.01)^2 = 1 + 0.1 + 0.004 = 1.104$, compared with the exact $1.1040808 \dots$

13. Binomial Expansion — Rational or Negative Index

For index n *not* a non-negative integer, the expansion is an *infinite series*:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

The series converges (and equals $(1 + x)^n$) only when $|x| < 1$. For $(a + x)^n$ with $a \neq 1$, factor out a first: $(a + x)^n = a^n \left(1 + \frac{x}{a}\right)^n$, valid for $|x/a| < 1$, i.e. $|x| < |a|$.

Worked Example 16 — First four terms of $(1 + x)^{-2}$

1. $n = -2$. Terms:
2. Term 0: 1
3. Term 1: $n \cdot x = (-2) \cdot x = -2x$

4. Term 2: $n(n-1)/2! \cdot x^2 = (-2)(-3)/2 \cdot x^2 = 3x^2$
5. Term 3: $n(n-1)(n-2)/3! \cdot x^3 = (-2)(-3)(-4)/6 \cdot x^3 = -4x^3$
6. $\therefore (1+x)^{-2} \approx 1 - 2x + 3x^2 - 4x^3 + \dots$, valid for $|x| < 1$.

Cross-check: differentiate the geometric series $1/(1-u) = 1 + u + u^2 + \dots$ or recall $1/(1+x)^2 = \sum_{k \geq 0} (-1)^k (k+1)x^k = 1 - 2x + 3x^2 - 4x^3 + \dots$ ✓

Exam tip — validity. When you write an expansion of $(1+x)^n$ for non-integer n , *always* state the range of validity ($|x| < 1$, or the appropriate scaled version). The "validity" mark is separate from the expansion marks on nearly every OCR paper.

14. Consolidated Summary and Exam Strategy

Chapter-wide checklist for Algebra and Functions.

- **Indices:** convert radicals to fractional powers before manipulating (e.g. $\sqrt[3]{x^2} = x^{2/3}$).
- **Surds:** always rationalise; keep surds simplified ($\sqrt{50} = 5\sqrt{2}$).
- **Quadratics:** factorise first, formula second, completing the square when you need the vertex/range/min-max.
- **Inequalities:** sketch or use a sign diagram; reverse the sign when multiplying/dividing by a negative.
- **Polynomial division / factor theorem:** test $\pm 1, \pm 2, \pm 3$ before larger divisors of the constant term.
- **Transformations:** inside the bracket = horizontal and reversed; outside = vertical and in the obvious direction.
- **Partial fractions:** match the factor structure exactly — repeated factors need multiple terms; irreducible quadratics need a linear numerator.
- **Functions:** always state the domain of an inverse. Check one-to-one before asserting invertibility.
- **Modulus:** two cases, or the graphical method — always verify candidate solutions against the original equation.

- **Binomial:** for integer n , finite expansion via Pascal's triangle; for rational/negative n , infinite series convergent for $|x| < 1$ — state validity.

Typical OCR examiner complaints (from mark reports).

- Leaving surds in denominators when a "simplified" answer is requested.
- Not stating the domain of f^{-1} .
- Omitting the validity $|x| < 1$ after a binomial expansion for non-integer index.
- Reversing the solution of a quadratic inequality (inside-vs-outside the roots confusion).
- Forgetting the repeated-factor term $B/(x - a)^2$ in a partial fraction setup.
- Accepting an extraneous "solution" to a modulus equation without checking that both sides have the same sign.